

Executive Summary

What problem am I trying to solve?

Do small language models internally encode a latent “reasoning mode” when producing Chain-of-Thought (CoT) answers, or are the differences between CoT and Direct prompting purely stylistic?

Why this is interesting

If a “reasoning mode” is detectable and controllable in activations, it could be directly safety-relevant: researchers could monitor when a model is reasoning step-by-step and potentially steer it. Probes and interventions provide a concrete way to test this hypothesis.

High-level takeaways

- Linear probes trained on GPT-2 separate CoT vs Direct final sentences very well (AUROC ≈ 0.95).
- Lexical baselines perform equally well, showing the effect is stylistic, not latent reasoning.
- Entropy is a weaker predictor, and sequence length is near chance.
- Probe signal localizes in mid-late layers.
- Steering attempts fail: the probe is *readable but not steerable*.
- Replication in distilgpt2 yields the same pattern.

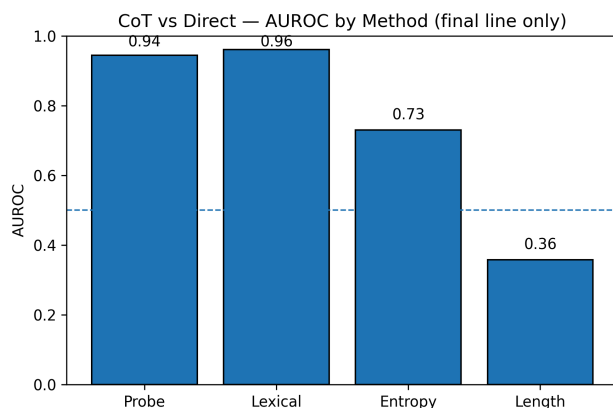
Key Experiments

1. Probe vs Baselines

Hypothesis: If reasoning mode is latent, probes should outperform surface baselines.

Result: Probe AUROC ≈ 0.95 ; lexical baseline ≈ 0.96 ; entropy weaker (0.73); length chance (0.36).

Takeaway: Separation is stylistic, not latent.

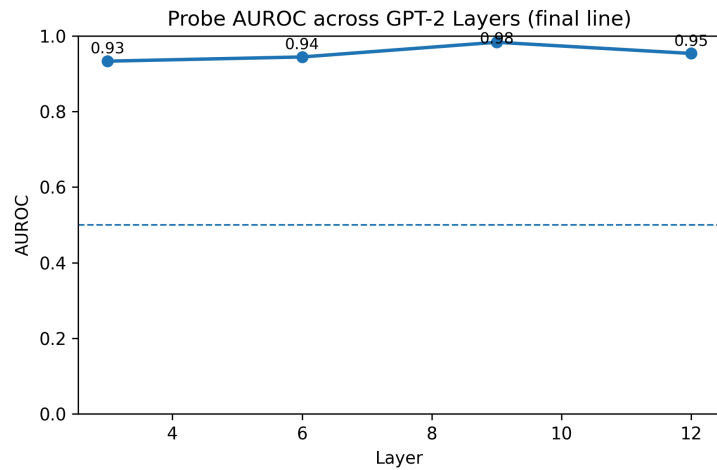


2. Layer-wise Localization

Hypothesis: If reasoning mode is encoded, probe accuracy should peak in specific layers.

Result: Performance peaks in mid-late layers (layer 9 for GPT-2).

Takeaway: Representation is strongest late in the forward pass, but reflects output style.

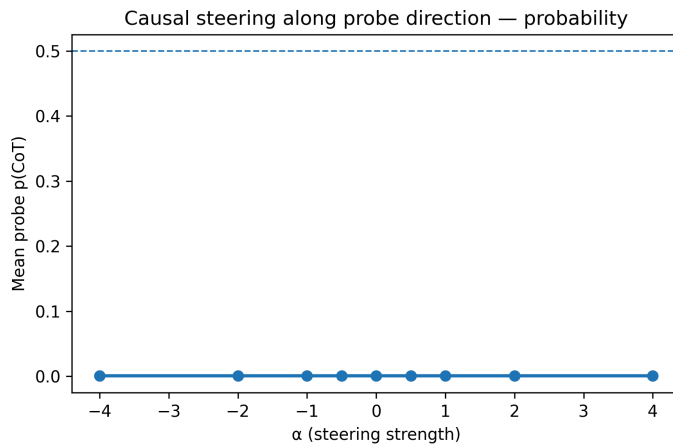


3. Causal Steering

Hypothesis: Injecting probe direction should shift behavior.

Result: Probe probabilities/logits flat across α sweep.

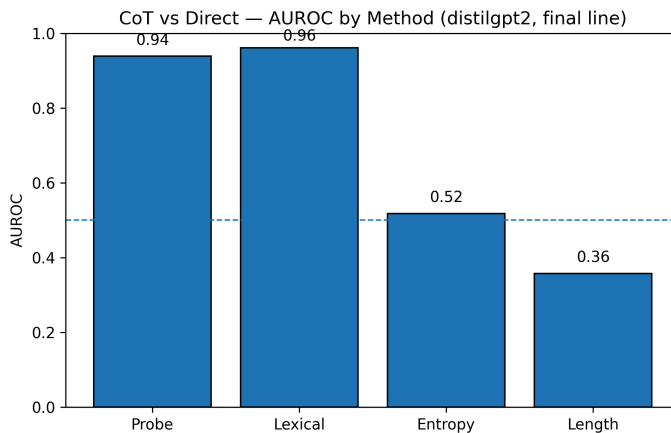
Takeaway: Representation is readable, not steerable.



4. Replication in distilgpt2

Result: Probe rivals Lexical again; entropy rivals chance.

Takeaway: Findings generalize across architectures.



Conclusion

Linear probes can monitor whether GPT-2 produces CoT or Direct answers, but the effect is largely explained by lexical style. Attempts at causal steering failed to change model behavior. This could reflect several factors:

- Surface correlation dominance: Final-line lexical features dominate the hidden states, so probes capture surface-level differences rather than deeper reasoning processes.
- Representation entanglement: If a “reasoning mode” exists, it may be distributed across many neurons or subspaces, not aligned with a single linear direction.
- Steering site limitations: Intervening at the residual stream of one layer may not be the right causal handle; other sites (e.g., MLP outprojections or earlier layers) might carry stronger causal leverage.
- Model scale: Small models like GPT-2 may not clearly separate reasoning from style; larger models might develop more distinct internal representations.

Thus, the results don’t necessarily mean models lack a reasoning mode, but they show that with small models and linear probes, what’s captured is surface-level, correlational signal rather than a causally controllable latent.

Main Body

Problem & Hypothesis

The motivating question was whether small language models (here GPT-2 and distilgpt2) encode a latent “reasoning mode” when producing Chain-of-Thought (CoT) outputs, distinct from surface stylistic differences.

- **Hypothesis:** If such a latent exists, linear probes should outperform simple baselines (like lexical or entropy features). Further, causal steering along the probe direction should shift model behavior.
- **Alternative hypothesis:** If probes perform comparably to surface baselines, the separation is explained by lexical style, not reasoning.

Data & Setup

Prompts and Outputs:

- I generated paired *Direct* and *CoT* answers for trivia and reasoning questions.
- For each example, I extracted the **final sentence/line** as the classification target. This choice reflected the intuition that the “answer” style is most diagnostic of reasoning vs directness.

Dataset Size: ~70 paired examples (~140 outputs total). Small but sufficient for proof-of-concept probing.

Models:

- GPT-2 (12 layers, hidden size 768).
- distilgpt2 (6 layers, hidden size 768) for replication.

Feature Extraction:

- Collected hidden activations at selected layers (GPT-2: 3, 6, 9, 12; distilgpt2: 2, 4, 6).
- Pooled across tokens in the final sentence.

Experiment 1: Probe vs Baselines

Hypothesis: If reasoning mode is latent, probes should beat surface baselines.

Method:

- Trained logistic regression probes on hidden states.
- Compared to three baselines:
 1. **Lexical (TF-IDF bag-of-words).**
 2. **Entropy** of next-token distribution before the final line.
 3. **Answer length.**
- Used GroupKFold CV by question (to prevent memorization).

Outcome:

- Probe AUROC is around 0.95.
- Lexical baseline nearly identical (0.96).
- Entropy weaker (0.73 GPT-2; 0.52 distilgpt2).
- Length ~chance (0.36).

Interpretation:

- The probe is highly predictive, but so is the lexical baseline.
- Strong evidence the probe is reading off *stylistic surface differences*, not hidden “reasoning.”
- Entropy provides some information (uncertainty higher in CoT), but not as strong.

Experiment 2: Layer-wise Localization

Hypothesis: If reasoning is represented internally, probes will peak in specific layers.

Method:

- Trained probes on GPT-2 layers 3, 6, 9, 12.
- Evaluated AUROC per layer.

Outcome:

- Probes improved with depth.
- Strongest performance at layer 9 (≈ 0.99 AUROC).
- distilgpt2 showed the same pattern, with layer 6 best.

Interpretation:

- CoT vs Direct separation signal is strongest in mid-late layers.
- But given the lexical baseline result, this localization likely reflects final-token style encoding, not reasoning mode.

Experiment 3: Causal Steering

Hypothesis: If the probe identifies a causal “reasoning direction,” injecting an alpha into hidden states should shift behavior.

Method:

- Took the probe weight vector at the best layer (9 for GPT-2).
- During generation, added $\pm\alpha \cdot w$ ($\alpha \in [-4, -2, -1, -0.5, 0, 0.5, 1, 2, 4]$) to the residual stream at final-line tokens.
- Measured changes in probe probability and logits.

Outcome:

- Both probability and logit curves are flat across α .

Interpretation:

- The probe captures correlational differences but does not control model behavior.
- Possible reasons:
 - Style features dominate the hidden states.
 - Any latent “reasoning mode” is distributed, not aligned with a single linear vector.
 - Residual stream at one layer may not be the right intervention site.
 - Model scale may be too small to develop a clear separation.

Experiment 4: Replication in distilgpt2

Hypothesis: Findings should generalize to related models

Outcome:

- distilgpt2 results identical: probe $\approx$ lexical, entropy weaker, length chance.

Interpretation:

- Confirms that separation is stylistic and not model-specific.

Discussion & Limitations

- **Takeaway:** Probes can separate CoT vs Direct answers, but baselines show this reflects style, not latent reasoning.
- **Limitations:** Small dataset, small models, only final-line features.
- **Alternative explanations:** Reasoning representations may exist but are (i) entangled across many features, (ii) stored elsewhere in the network, or (iii) clearer in larger models.
- **Next steps:** Scale to larger models, probe intermediate CoT tokens, test other intervention sites (e.g., MLP projections), or apply SAEs to disentangle features.

Toggl:

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write up	● mats application	9:08 PM - 10:31 PM	1:23:19
general work	● mats application	12:42 PM - 9:08 PM	8:26:03
☐ Yesterday			5:04:13
brainstorming and planning	● mats application	5:15 PM - 10:19 PM	5:04:13